

Comprehensive Research Analysis: IIT Bombay

Department of Chemistry - Current and Recent Research Activities

Based on extensive analysis of faculty profiles, recent publications, and current research projects spanning 2020-2024, the Department of Chemistry at IIT Bombay represents one of India's most comprehensive and cutting-edge chemistry research environments, encompassing theoretical and experimental work across multiple frontiers of modern chemistry.

Advanced Quantum Chemistry and Computational Methods

Prof. Achintya Kumar Dutta leads groundbreaking theoretical chemistry research developing lower scaling wave-function based methods for quantum chemistry applications. His recent breakthrough work focuses on electron attachment to DNA and the protective role of amino acids, examining electron attachment to wobble base pairs, and developing approximate EOM-CCSD methods for ionization potential and electron affinity calculations of genetic material subunits. His group's cutting-edge contributions include similarity transformed second-order approximate coupled cluster methods for excited states, pair natural orbital equation-of-motion coupled cluster methods for core binding energies, and simulation of X-ray spectroscopy for molecular systems. As a lead author of the popular ORCA quantum chemistry software package, his methodological developments directly impact the global computational chemistry community.^{[85][86]}

Prof. Rahul Maitra specializes in many-body quantum chemistry with focus on developing novel electronic structure methodologies for strongly correlated molecular systems. His recent revolutionary work includes formulation of dressed coupled cluster methods with implicit triple excitations for hydrogen-bonded systems, accelerating coupled cluster calculations using nonlinear dynamics and supervised machine learning, and developing approximate coupled cluster theories via nonlinear dynamics and synergetics with adiabatic decoupling conditions. His group's innovative approaches include stability analysis of double similarity transformed coupled cluster theory and geometric dependence studies of molecular dipole polarizability in water, addressing higher-order electron correlation, basis set incompleteness errors, core electron effects, and zero-point vibrational contributions.^{[87][88]}

Fluorescence Spectroscopy and Ultrafast Dynamics

Prof. Anindya Datta operates one of India's most advanced ultrafast dynamics research programs, combining state-of-the-art spectroscopic techniques with fundamental investigations of molecular processes. His recent groundbreaking research includes microscopic perspectives of synergy between localized surface plasmon resonance and disruption of dye aggregates in metal nanoparticle-enhanced fluorescence, noninvasive contactless characterization of electronic properties at semiconductor/dielectric interfaces using optical second-harmonic generation, and single particle dynamics of carrier replenishment in quantum dots. His group's cutting-edge work encompasses disruption of aggregates of zinc complexes by surfactants through ensemble and single molecule fluorescence spectroscopy, photoactivation and suppression of Auger recombination in quantum dots, and mechanistic insights into selective sensing of heavy metals using photoluminescent quantum dots. The laboratory houses sophisticated instrumentation including fluorescence upconversion systems, pump-probe spectrophotometers, time-correlated single photon counting setups, and custom-built ultrafast laser systems.^{[89][90][91][92]}

Advanced Polymer Chemistry and Flow Chemistry

Prof. Anil Kumar leads India's pioneering research in continuous flow chemistry and conducting polymers, establishing the country's first state-of-the-art continuous flow chemistry laboratory. His revolutionary contributions include development of processable conducting polymers for printable molecular electronics, high-throughput template-free continuous flow synthesis of polyaniline nanofibers, and exceptional photoconductivity in polymer composites through in-situ encapsulation of quantum dots. His group's industrial collaborations span major chemical and pharmaceutical companies including Gland Pharma, Jubilant, Merck/Aldrich, Reliance Industries, Sun Pharma, and Syngenta, demonstrating the practical impact of his research. Recent breakthrough work includes role of molecular and interchain ordering in organic solar cells, ultra-high mobility in defect-free conducting polymers through supramolecular alignment, and development of ITO-free flexible electrochromic devices. His laboratory capabilities include highly corrosive reagent processing using silicon carbide reactors, high-temperature solution phase chemistry up to 500°C, continuous crystallization systems, and comprehensive process analytical technology (PAT) tools including flow IR, flow NMR, and real-time reaction monitoring.^{[93][94]}

Carbohydrate Chemistry and Glycomics

Prof. Suvarn S. Kulkarni pioneered one-pot methodologies for glycomics and synthesis of cell-surface carbohydrates with applications in vaccine development and cancer research. His recent groundbreaking achievements include total syntheses of conjugation-ready trisaccharide repeating units of *Pseudomonas*

aeruginosa O11 and Staphylococcus aureus Type 5 capsular polysaccharides for vaccine development, investigation of virulence-associated glycolipids with distinct conformational attributes affecting host plasma membrane organization, and development of novel protecting groups for oligosaccharide synthesis. His work on total synthesis of lipid-anchor-attached core trisaccharides of lipoteichoic acids from pathogenic bacteria and assembling glycan-charged dolichol phosphates for N-glycosylation pathway intermediates represents cutting-edge contributions to understanding bacterial infections and cellular processes. His research directly addresses infectious diseases including tuberculosis, AIDS, malaria, stomach infections, and various cancers through rational design of carbohydrate-based therapeutics.^{[95][96][97]}

Electrochemical Energy Conversion and Storage

Prof. Srinivasan Ramakrishnan conducts molecular-level design of interfacial electrochemical processes in energy conversion systems. His pioneering research includes extended interfacial stability through simple acid rinsing in Li-rich oxide cathode materials, definition of redox centers in lithium intercalation reactions, and mapping free energy regimes in electrocatalytic reductions to screen transition metal-based catalysts. His experience as Electrochemical Engineering Lead at Neuralink Corporation brings unique perspectives to brain-machine interface applications, while his work on initiation of electrochemical CO₂ reduction by singly reduced ruthenium complexes and multi-electron transfer at cobalt centers addresses critical challenges in sustainable energy conversion. His current projects span materials and interface engineering for long cycle life sodium-sulfur batteries, development of Co-free/less high voltage cathode materials for Li-ion batteries, and fluoride-rich interlayers for anodeless solid-state batteries.^{[98][99][100][101]}

Organic Synthesis and Natural Products

Prof. Santosh J. Gharpure leads comprehensive organic synthesis research spanning natural product synthesis, asymmetric synthesis, and catalysis. His recent innovative contributions include design and expeditious synthesis of quinoline-pyrene based ratiometric fluorescent probes for targeting lysosomal pH, expeditious diastereoselective synthesis of medium ring heterocycle-fused chromenes via tandem hydroalkoxylation-cycloaddition reactions, and TMSOTf mediated reductive hydroamination for stereoselective synthesis of nitrogen heterocycles. His cascade radical cyclization strategies for divergent synthesis of substituted furans and dihydrofurans, transition-metal acetate-promoted intramolecular nitrene insertion reactions, and development of alkynyl hydroxylamines as key precursors for 1,2-N/O heterocycles represent cutting-edge methodological advances. As Rasiklal Hemani Fragrance and Flavour

Chair Professor and Professor In-Charge of SINE (IIT Bombay's technology incubator), he bridges fundamental research with industrial applications.^{[102][103][104][105]}

Molecular Magnetism and Computational Chemistry

Prof. G. Rajaraman operates one of the world's leading computational molecular magnetism research programs, combining density functional theory, ab initio methods, and experimental validation. His groundbreaking research includes synthetic strategies for switching single ion anisotropy in tetrahedral cobalt complexes, comprehensive studies of magnetic anisotropy mechanisms in lanthanide single-ion magnets, and exploration of unusual oxidation states for unprecedented SMM performance. His group's cutting-edge work encompasses design of single-molecule toroics using lanthanide triangle complexes, theoretical exploration of transition metal and lanthanide-based SMMs, and quantum chemical investigation of magnetic exchange, anisotropy, and excited state energetics. Recent breakthrough contributions include lanthanide complexes as molecular dopants for air-stable n-type graphene logic inverters and comprehensive studies of what controls magnetic anisotropy in tetrahedral cobalt single-ion magnets.^{[106][107][108][109]}

Peptide Chemistry and Ion Channel Mimetics

Prof. Nandita Madhavan develops minimalistic peptide scaffolds that mimic protein function, particularly focusing on transmembrane ion channels. Her innovative research includes engineering biomimetic molecules using amino acids as building blocks, exploiting functional group diversity and hydrogen bonding capability of peptides to develop three-dimensional supramolecular structures. Recent groundbreaking work encompasses macrocyclic transmembrane anion transporters via one-pot condensation reactions, low molecular weight di- to tetrapeptide transmembrane cation transporters, and development of phosphorus pentoxide for amide and peptide bond formation with minimal by-products. Her group's cutting-edge contributions include rink-amide soluble supports for high purity peptide synthesis, aromatic dipeptides as anion transporters, and anion selective, pH responsive macrocyclic systems with applications in cellular transport and molecular switches.^{[110][111][112]}

Metal-Organic Frameworks and Porous Materials

Prof. Sanjog S. Nagarkar focuses on designed assembly of molecular materials using dynamic coordination bonds for coordination cages and coordination polymers. His recent breakthrough research includes CeVO₄ nanosheets as bifunctional electrocatalysts for glycerol oxidation and hydrogen peroxide

sensing, template-free molten salt synthesis of Sr-doped LaCrO_3 1D nanorods with enhanced electrical transport properties, and comprehensive reviews of metal-organic frameworks for hydrogen gas storage applications. His innovative work encompasses proton conducting metal-organic frameworks and their applications, role of MOFs in hydrogen gas storage with critical analysis of challenges and opportunities, and coordination polymer glasses with controllable network sizes. His research addresses both scientific understanding of framework materials and industrially relevant applications in chemical storage, separation, sustainable energy conversion, and biomimetic catalysis.^{[113][114][115]}

Biophysical Chemistry and Chemical Biology

Prof. Shobhna Kapoor specializes in molecular biophysics and chemical biology with focus on lipids and lipid-induced signaling pathways. Her cutting-edge research encompasses the emerging world of membrane vesicles and their functional relevance, switching between cellular blebbing and lamellipodia depending on non-muscle myosin II activity, and ligand-induced copper transport restriction in cancer and mycobacterial growth. Her innovative work on lipidomic landscapes in cancer provides actionable insights for membrane-based therapy and diagnoses, while her development of plug-and-select ion channel scaffolds advances understanding of cellular transport mechanisms.

Computational Materials Chemistry

Prof. M.S. Balakrishna designs multifunctional, multidentate phosphorus, nitrogen and silicon-based ligands with applications in organometallic chemistry. His recent breakthrough work includes cationic and neutral palladium and platinum pincer complexes, synthesis and photophysical properties of heavier pnictogens, and novel 1,2,3-triazolyl phosphines with pyridyl functionality for coinage metal complexes and catalytic applications. His copper complexes of amide functionalized bisphosphines demonstrate proximity-enhanced metal-ligand cooperativity with catalytic advantages in $\text{C}(\text{sp}^3)\text{-H}$ bond activation.

Nanotechnology and Carbon Materials

Prof. Chandramouli Subramaniam investigates structure-property relationships of carbon nanotubes, graphene, and related doped nanocarbons including nitrogen and boron-doped systems. His revolutionary contributions include plasmonic synergism in tailored metal-carbon interfaces for real-time single molecular level detection of environmental pollutants, scalable high-throughput photothermal water disinfection with negligible CO_2 footprint, and magneto-electrocatalytic overall water splitting achieved at

unprecedented low voltages. His phonon-engineered hard-carbon nanoflorets achieve rapid and efficient solar-thermal based water evaporation with applications in sustainable water treatment and space heating.

Soft Materials and Protein Engineering

Prof. Kamendra P. Sharma focuses on design and development of multi-functional nanomaterials via non-covalent interactions combining proteins, polymers, surfactants, nanoparticles, and liquid crystals. His groundbreaking research includes intrinsic elasticity of three-dimensional macroporous scaffolds governing kinetics of biomimetic reactions, amyloid-like aggregation in native proteins and suppression in bio-conjugate counterparts, and viscoelastic liquids from regenerated silk fibroin as writeable and shapeable materials. His innovative work on ultraslow biological water-like dynamics in waterless liquid proteins and light-responsive self-assembled microstructures of branched polyethyleneimine provides fundamental insights into protein behavior and materials design.

Advanced Spectroscopy and Materials Characterization

Prof. Deepti Kalsi specializes in surface organometallic chemistry, photocatalysis, and tandem catalysis. Her cutting-edge research includes one-pot multicomponent synthesis using catalytic systems composed of ruthenium nanoparticles on modified silica supports, selectivity control in hydrogenation through adaptive catalysis using responsive supports, and ruthenium-catalyzed selective hydro-boronolysis of ethers. Her innovative approaches to transition metal-catalyzed reactions and development of heterogeneous catalytic systems address key challenges in sustainable chemical synthesis.